

Computer Vision Final Project Proposal: GeoGuessr AI using DINOv3

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1. Problem Statement

This project aims to train a lightweight prediction head on top of the **DINOv3** vision transformer to predict geographic coordinates (latitude and longitude) from street-view images for automated GeoGuessr gameplay. DINOv3 is the latest large-scale vision transformer that has demonstrated state-of-the-art performance across various vision tasks, often surpassing many fine-tuned and domain-specific models. This project investigates whether DINOv3's powerful learned representations can compete with specialized geolocation models like **PIGEON**, which was specifically designed for this task and achieved top 0.01% human player performance.

The core research question is: **Can a general-purpose vision foundation model (DINOv3) match or exceed specialized geolocation models when trained with a lightweight prediction head?**

2. Problem Relevance

Practical Applications

While geolocation from images is entertaining in the context of GeoGuessr gameplay, it has significant real-world applications:

- **Crime Investigation:** Identifying locations from surveillance footage, social media images, or evidence photos
- **Search and Rescue:** Locating individuals from photos or videos when GPS data is unavailable
- **Digital Forensics:** Verifying the authenticity and origin of digital content
- **Content Moderation:** Identifying geographic origins of problematic online content
- **Wildlife Conservation:** Tracking animal locations from camera trap images

Academic Interest

This project also explores the **transfer learning capabilities** of large foundation models, testing whether general-purpose visual representations can be efficiently adapted to specialized tasks with minimal additional training.

3. Prior Art and Related Work

Existing Geolocation Models

PIGEON (Stanford, 2024): The current state-of-the-art geolocation model achieved 92% country-level accuracy and >40% accuracy within 25km globally. It uses semantic geocell creation and multi-task contrastive pretraining, ranking in the top 0.01% of human GeoGuessr players.

OpenAI's o3 (2025): Recently achieved superhuman performance with 23,179 points vs. Master I-ranked human's 22,054 points by combining vision capabilities with chain-of-thought reasoning.

GeoCLIP (2023): Introduced hierarchical location encoding using random Fourier features, achieving 74.1% continent-level accuracy through CLIP-inspired alignment between locations and images.

GAEA (2025): A conversational geolocation model achieving 66.06% average accuracy with conversational reasoning capabilities.

Foundation Models

DINOv3: A self-supervised vision transformer trained on diverse visual data that has demonstrated exceptional transfer learning performance across multiple downstream tasks without task-specific fine-tuning.

4. Proposed Method

Model Architecture

The proposed approach leverages DINOv3's frozen pre-trained backbone with a trainable prediction head:

1. **Feature Extraction:** Use pre-trained **DINOv3-ViT** (e.g., ViT-L/14 or ViT-g/14) as a frozen feature extractor
2. **Prediction Head:** Train a lightweight head on top, with two architecture options:
 - **Option A:** Simple linear layer(s) for direct coordinate regression
 - **Option B:** Additional transformer layers for spatial reasoning, following DINOv3's text-alignment example

Training Strategy

Dataset: OpenStreetView-5M (OSV5M) dataset or Google Street View datasets from Kaggle

- OSV5M contains 5.1M geo-referenced street view images from 225 countries
- Alternative: Start with smaller Kaggle datasets (10,000-25,000 images) for prototyping

Training Approach:

- Freeze DINOv3 backbone weights to leverage pre-trained representations
- Train only the prediction head to minimize computational requirements
- Use geocell classification (similar to PIGEON) rather than direct regression to avoid "landing in oceans"
- Alternatively, explore hybrid approach: coarse classification + fine coordinate regression

Expected Outcomes

- **Success:** Achieve country-level accuracy >80% and demonstrate that DINOv3's representations are effective for geolocation
 - **Baseline:** Match or approach PIGEON's performance
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References

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